

E/22/340

MaxEntropy Interference assignment.

ME2050-Introduction to statical learning

1. The Partition Function (Z)

The discrete energy levels of the quantum harmonic oscillator are given by $E_j = \hbar\omega \left(j + \frac{1}{2}\right)$ for $j = 0, 1, 2, \dots$. The canonical partition function is defined as:

$$Z(T) = \sum_{j=0}^{\infty} e^{-\frac{E_j}{k_B T}} = \sum_{j=0}^{\infty} e^{-\frac{\hbar\omega \left(j + \frac{1}{2}\right)}{k_B T}}$$

Factoring out the ground-state constant term ($j = 0$ component):

$$Z(T) = e^{-\frac{\hbar\omega}{2k_B T}} \sum_{j=0}^{\infty} \left(e^{-\frac{\hbar\omega}{k_B T}} \right)^j$$

Let $r = e^{-\frac{\hbar\omega}{k_B T}}$. Because $\hbar, \omega, k_B, T > 0$, the ratio satisfies $0 < r < 1$. The summation forms a convergent infinite geometric series following the standard identity $\sum_{j=0}^{\infty} r^j = \frac{1}{1-r}$:

$$Z(T) = \frac{e^{-\frac{\hbar\omega}{2k_B T}}}{1 - e^{-\frac{\hbar\omega}{k_B T}}}$$

2. Thermodynamic Potentials

Let $\beta = \frac{1}{k_B T}$. Expressing Z in terms of β :

$$Z = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1 - e^{-\beta\hbar\omega}}$$

Taking the natural logarithm:

$$\ln Z = -\frac{\beta\hbar\omega}{2} - \ln(1 - e^{-\beta\hbar\omega})$$

- **Helmholtz Free Energy ($F(T)$):**

$$F(T) = -k_B T \ln Z = \frac{\hbar\omega}{2} + k_B T \ln \left(1 - e^{-\frac{\hbar\omega}{k_B T}} \right)$$

- **Internal Energy ($U(T)$):**

$$U(T) = -\frac{\partial \ln Z}{\partial \beta} = -\frac{\partial}{\partial \beta} \left[-\frac{\beta \hbar \omega}{2} - \ln(1 - e^{-\beta \hbar \omega}) \right]$$

$$U(T) = \frac{\hbar \omega}{2} + \frac{1}{1 - e^{-\beta \hbar \omega}} \cdot (-e^{-\beta \hbar \omega}) \cdot (-\hbar \omega)$$

$$U(T) = \frac{\hbar \omega}{2} + \frac{\hbar \omega e^{-\beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}} = \frac{\hbar \omega}{2} + \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1}$$

Substituting back $\beta = \frac{1}{k_B T}$:

$$U(T) = \frac{\hbar \omega}{2} + \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1}$$

3. Entropy (S)

Using the thermodynamic relation $S = -\left(\frac{\partial F}{\partial T}\right)$:

$$S = -\frac{\partial}{\partial T} \left[\frac{\hbar \omega}{2} + k_B T \ln \left(1 - e^{-\frac{\hbar \omega}{k_B T}} \right) \right]$$

Applying the product rule:

$$S = -k_B \ln \left(1 - e^{-\frac{\hbar \omega}{k_B T}} \right) - k_B T \cdot \frac{1}{1 - e^{-\frac{\hbar \omega}{k_B T}}} \cdot \left(-e^{-\frac{\hbar \omega}{k_B T}} \right) \cdot \left(\frac{\hbar \omega}{k_B T^2} \right)$$

$$S(T) = k_B \left[\frac{\frac{\hbar \omega}{k_B T}}{e^{\frac{\hbar \omega}{k_B T}} - 1} - \ln \left(1 - e^{-\frac{\hbar \omega}{k_B T}} \right) \right]$$

Verification of $U = F + TS$:

$$F + TS = \left[\frac{\hbar \omega}{2} + k_B T \ln \left(1 - e^{-\frac{\hbar \omega}{k_B T}} \right) \right] + k_B T \left[\frac{\frac{\hbar \omega}{k_B T}}{e^{\frac{\hbar \omega}{k_B T}} - 1} - \ln \left(1 - e^{-\frac{\hbar \omega}{k_B T}} \right) \right]$$

The log terms cancel out perfectly, leaving:

$$F + TS = \frac{\hbar \omega}{2} + \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} = U(T)$$

4. Heat Capacity and Hyperbolic Form

The constant-volume heat capacity is $C_V = \frac{\partial U}{\partial T}$. Differentiating our expression for $U(T)$:

$$C_V = \frac{\partial}{\partial T} \left[\frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_B T}} - 1} \right] = \hbar\omega \cdot \frac{-1}{\left(e^{\frac{\hbar\omega}{k_B T}} - 1\right)^2} \cdot e^{\frac{\hbar\omega}{k_B T}} \cdot \left(-\frac{\hbar\omega}{k_B T^2}\right)$$

$$C_V = k_B \left(\frac{\hbar\omega}{k_B T}\right)^2 \frac{e^{\frac{\hbar\omega}{k_B T}}}{\left(e^{\frac{\hbar\omega}{k_B T}} - 1\right)^2}$$

To express this in hyperbolic form, multiply the numerator and denominator by $e^{-\frac{\hbar\omega}{k_B T}}$:

$$C_V = k_B \left(\frac{\hbar\omega}{k_B T}\right)^2 \frac{1}{\left(e^{\frac{\hbar\omega}{2k_B T}} - e^{-\frac{\hbar\omega}{2k_B T}}\right)^2}$$

Using the identity $\sinh(x) = \frac{e^x - e^{-x}}{2} \Rightarrow e^x - e^{-x} = 2 \sinh(x)$, let $x = \frac{\hbar\omega}{2k_B T}$:

$$C_V = k_B \left(\frac{\hbar\omega}{k_B T}\right)^2 \frac{1}{\left(2 \sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right)^2} = k_B \left(\frac{\hbar\omega}{2k_B T}\right)^2 \frac{1}{\sinh^2\left(\frac{\hbar\omega}{2k_B T}\right)}$$

5. Asymptotic Behavior (Discussion)

In the high-temperature limit where $k_B T \gg \hbar\omega$, the dimensionless variable $x = \frac{\hbar\omega}{2k_B T} \rightarrow 0$.

Using the Taylor series expansion for small values, $\sinh(x) \approx x$. Substituting this approximation into the hyperbolic capacity expression:

$$C_V \approx k_B (x)^2 \cdot \frac{1}{(x)^2} = k_B$$

Yes, this matches the classical equipartition theorem prediction. A classical one-dimensional harmonic oscillator has two degrees of freedom (kinetic energy and quadratic potential energy), each contributing $\frac{1}{2} k_B$ to the heat capacity, resulting in $C_V = k_B$.

2. Maximum Entropy Distribution under Mean and Variance Constraints

We aim to maximize the continuous differential Shannon entropy:

$$H(p) = - \int_{-\infty}^{\infty} p(x) \ln p(x) dx$$

Subject to the following strict constraints:

1. $\int_{-\infty}^{\infty} p(x) dx = 1$ (Normalization)
2. $\int_{-\infty}^{\infty} x p(x) dx = \mu$ (Mean constraint)
3. $\int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx = \sigma^2$ (Variance constraint)

Using the optimization framework of Lagrange Multipliers, we formulate the functional:

$$\begin{aligned} \mathcal{L}(p, \lambda_0, \lambda_1, \lambda_2) &= - \int p(x) \ln p(x) dx - \lambda_0 \left(\int p(x) dx - 1 \right) - \lambda_1 \left(\int x p(x) dx - \mu \right) \\ &\quad - \lambda_2 \left(\int (x - \mu)^2 p(x) dx - \sigma^2 \right) \end{aligned}$$

Taking the functional derivative with respect to $p(x)$ and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial p(x)} = -\ln p(x) - 1 - \lambda_0 - \lambda_1 x - \lambda_2 (x - \mu)^2 = 0$$

$$\ln p(x) = -(1 + \lambda_0) - \lambda_1 x - \lambda_2 (x - \mu)^2$$

$$p(x) = \exp(-(1 + \lambda_0)) \exp(-\lambda_1 x - \lambda_2 (x - \mu)^2)$$

To satisfy the centralized properties of the mean constraint μ , λ_1 must equal 0. Let $C = \exp(-(1 + \lambda_0))$ serve as the normalization constant:

$$p(x) = C \exp(-\lambda_2 (x - \mu)^2)$$

Using the Gaussian integration identity $\int_{-\infty}^{\infty} e^{-a(x-\mu)^2} dx = \sqrt{\frac{\pi}{a}}$, normalization requires:

$$C \sqrt{\frac{\pi}{\lambda_2}} = 1 \Rightarrow C = \sqrt{\frac{\lambda_2}{\pi}}$$

Next, checking the variance constraint:

$$\int_{-\infty}^{\infty} \sqrt{\frac{\lambda_2}{\pi}} (x - \mu)^2 e^{-\lambda_2 (x - \mu)^2} dx = \sigma^2$$

Using the standard definite integral calculus result $\int_{-\infty}^{\infty} y^2 e^{-ay^2} dy = \frac{1}{2a} \sqrt{\frac{\pi}{a}}$:

$$\sqrt{\frac{\lambda_2}{\pi}} \cdot \frac{1}{2\lambda_2} \sqrt{\frac{\pi}{\lambda_2}} = \sigma^2 \Rightarrow \frac{1}{2\lambda_2} = \sigma^2 \Rightarrow \lambda_2 = \frac{1}{2\sigma^2}$$

Substituting λ_2 back into C yields $C = \frac{1}{\sqrt{2\pi\sigma^2}}$. This leaves us with the exact analytical formulation of the normal distribution:

$$X \sim \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Under the Maximum Entropy principle for discrete conditional distributions, the optimization naturally results in a Gibbs distribution form. For a binary target output $Y \in \{0,1\}$ (where $Y = 1$ designates routing to Technical Support), the conditional probability is characterized by log-linear parameters:

$$P(Y = 1 | \mathbf{x}) = \frac{e^{\theta^T \mathbf{x}}}{1 + e^{\theta^T \mathbf{x}}} = \frac{1}{1 + e^{-\theta^T \mathbf{x}}}$$

Where $\mathbf{x} = (x_1, x_2, x_3)^T$ represents the feature vector and $\theta = (\theta_1, \theta_2, \theta_3)^T$ represents the weight vectors associated with the historical system constraints. The observed joint data expectation values provided are:

- $\mathbb{E}[x_1 \cdot \mathbb{I}(Y = 1)] = \sum_{\mathbf{x}} P(\mathbf{x}) \cdot x_1 \cdot P(Y = 1 | \mathbf{x}) = 0.05$
- $\mathbb{E}[x_2 \cdot \mathbb{I}(Y = 1)] = \sum_{\mathbf{x}} P(\mathbf{x}) \cdot x_2 \cdot P(Y = 1 | \mathbf{x}) = 0.15$
- $\mathbb{E}[x_3 \cdot \mathbb{I}(Y = 1)] = \sum_{\mathbf{x}} P(\mathbf{x}) \cdot x_3 \cdot P(Y = 1 | \mathbf{x}) = 0.16$

We expand these expectations across our active historical operational scenarios:

- **Scenario A:** $\mathbf{x}_A = (1,0,0)^T \Rightarrow P(\mathbf{x}_A) = 0.40$
- **Scenario B:** $\mathbf{x}_B = (0,1,0)^T \Rightarrow P(\mathbf{x}_B) = 0.25$
- **Scenario C:** $\mathbf{x}_C = (0,0,1)^T \Rightarrow P(\mathbf{x}_C) = 0.05$
- **Scenario D:** $\mathbf{x}_D = (0,1,1)^T \Rightarrow P(\mathbf{x}_D) = 0.15$
- **Scenario E:** $\mathbf{x}_E = (1,1,0)^T \Rightarrow P(\mathbf{x}_E) = 0.05$

Let $P_{Y=1}(\mathbf{x}) = P(Y = 1 | \mathbf{x})$. Expanding the expectation constraints gives us a system of non-linear algebraic equations:

1. **Constraint 1 (x_1):**

$$P(\mathbf{x}_A) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_A) + P(\mathbf{x}_E) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_E) = 0.05$$

$$0.40P_{Y=1}(1,0,0) + 0.05P_{Y=1}(1,1,0) = 0.05$$

2. Constraint 2 (x_2):

$$P(\mathbf{x}_B) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_B) + P(\mathbf{x}_D) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_D) + P(\mathbf{x}_E) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_E) = 0.15$$

$$0.25P_{Y=1}(0,1,0) + 0.15P_{Y=1}(0,1,1) + 0.05P_{Y=1}(1,1,0) = 0.15$$

3. Constraint 3 (x_3):

$$P(\mathbf{x}_C) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_C) + P(\mathbf{x}_D) \cdot 1 \cdot P_{Y=1}(\mathbf{x}_D) = 0.16$$

$$0.05P_{Y=1}(0,0,1) + 0.15P_{Y=1}(0,1,1) = 0.16$$

By mapping the conditional probability equations directly to the parameters via logistic weights, $P_{Y=1}(\mathbf{x}) = \frac{1}{1+e^{-\theta^T \mathbf{x}}}$, these constraints are fed into an optimization solver accurately determine the parameter values θ .

We seek the risk-neutral probability density function $p(S_T)$ over $S_T \in (0, \infty)$ that maximizes the differential entropy:

$$H(p) = - \int_0^\infty p(S_T) \ln p(S_T) dS_T$$

Subject to the following observed market exchange constraints:

1. $\int_0^\infty p(S_T) dS_T = 1$
2. $\mathbb{E}[S_T] = \int_0^\infty S_T p(S_T) dS_T = 102$
3. $\mathbb{E}[S_T^2] = \int_0^\infty S_T^2 p(S_T) dS_T = 10,600$
4. $\mathbb{E}[\ln(S_T)] = \int_0^\infty \ln(S_T) p(S_T) dS_T = 4.60$

Formulating the objective function using Lagrange Multipliers ($\lambda_0, \lambda_1, \lambda_2, \lambda_3$):

$$\begin{aligned} \mathcal{L} = & - \int_0^\infty p(S_T) \ln p(S_T) dS_T - \lambda_0 \left(\int_0^\infty p(S_T) dS_T - 1 \right) - \lambda_1 \left(\int_0^\infty S_T p(S_T) dS_T - 102 \right) \\ & - \lambda_2 \left(\int_0^\infty S_T^2 p(S_T) dS_T - 10,600 \right) - \lambda_3 \left(\int_0^\infty \ln(S_T) p(S_T) dS_T - 4.60 \right) \end{aligned}$$

Differentiating with respect to $p(S_T)$ and setting to zero:

$$\frac{\partial \mathcal{L}}{\partial p(S_T)} = -\ln p(S_T) - 1 - \lambda_0 - \lambda_1 S_T - \lambda_2 S_T^2 - \lambda_3 \ln(S_T) = 0$$

$$\ln p(S_T) = -(1 + \lambda_0) - \lambda_1 S_T - \lambda_2 S_T^2 - \lambda_3 \ln(S_T)$$

Exponentiating both sides to isolate $p(S_T)$:

$$p(S_T) = e^{-(1+\lambda_0)} e^{-\lambda_1 S_T - \lambda_2 S_T^2} e^{-\lambda_3 \ln(S_T)}$$

Let $C = e^{-(1+\lambda_0)}$ be the normalizing constant. Since $e^{-\lambda_3 \ln(S_T)} = S_T^{-\lambda_3}$, we can simplify the equation to:

$$p(S_T) = C S_T^{-\lambda_3} \exp(-\lambda_1 S_T - \lambda_2 S_T^2)$$